

Dataset Handoff — Forensic Activity Reconstruction

Project: Development of an enhanced LLM-based digital forensic tool for mobile phone activity reconstruction

Source device: Google Pixel 3, Android 9 (Pie), build PQ2A.190205.001 Logical extraction (Axiom Process), decrypted /data partition Public corpus: Digital Corpora, image created by Joshua Hickman Population period: 2019-02-13 to 2019-04-06 (~7 weeks)

1. What you are receiving

File	Contents
train.jsonl	28 labelled chat-format pairs
unlabelled.jsonl	52 event windows, no targets — inference/test set
ground_truth.csv	322 documented events (the answer key)
events.jsonl	30,609 extracted events with full provenance
evaluation_report.txt	Baseline extraction metrics

Format

train.jsonl uses standard chat format, one JSON object per line:

```
{  
  "messages": [  
    {  
      "role": "system",  
      "content": "You are a digital forensic analyst..."  
    },  
    {  
      "role": "user",  
      "content": "Date: 2019-04-05\nEXTRACTED ARTIFACTS...\nPROVENANCE..."  
    },  
    {  
      "role": "assistant",  
      "content": "Reconstructed activity with artifact support:\n- 11:37 - Phone: Outgoing Call 1:44 [EVT-a3f9c21b7e04]..."  
    }  
  ]  
}
```

One pair per calendar day. Input is that day’s artifacts plus a provenance block; target is the correct reconstruction with citations.

2. Volume — read this first

28 labelled pairs is small. It constrains method choice:

- **LoRA / QLoRA** — viable at this scale, and the recommended approach
- **Full fine-tuning** — not viable, will overfit

- **Few-shot prompting** — a legitimate fallback; the same data works as in-context examples with no training at all

Prompts are long (a busy day carries 60–70 events plus provenance), so sequence length matters more than example count. Expect 4k–8k tokens per example.

If more data is needed, two options: 1. Re-window at session granularity instead of daily (3–5x more pairs) 2. Add a second Hickman image (Android 10, 11) — also gives a held-out test set

3. The behaviour being trained

The goal is a model sharper than a general LLM on forensic reasoning specifically. Four behaviours are encoded in the targets:

a) Citation discipline. Every claim ends with [EVT-xxxxxxxxxxxx]. IDs resolve to artifact :: table :: row_id :: raw_timestamp. An uncited claim is a hallucination by construction — verification is mechanical, not a matter of trusting the output.

b) Absence as evidence. Targets state explicitly when documented activity has no artifact support, and why. A general model says “no Signal activity found.” The target says “Signal was used; its store is encrypted and not recoverable from a logical extraction.”

c) Residue implies history. WeChat retains only cookies. The correct reading is *installed, used, removed* — not *never used*.

d) Timestamps can lie. 181 extracted events predate the device wipe. They are server-synced content carrying original creation times. Targets flag this rather than reporting activity before the device existed.

4. Baseline extraction performance

Category	GT events	Matched	Rate
DIRECT (recoverable)	157	107	68.2%
INDIRECT_ONLY	77	0	0.0%
PACKAGE_ABSENT	38	0	0.0%
NO_SQLITE	43	0	0.0%
ENCRYPTED	6	0	0.0%
RESIDUE_ONLY	1	0	0.0%

Timestamp accuracy on matched events: **84.1% exact**, mean delta 0.36 min, across five distinct epoch formats.

Strong per-app coverage: Chrome, Instagram, Phone, Messages, GalleryVault 100%; WhatsApp 88.9%; Snapchat / Telegram / Kik 87.5%.

Zero coverage: Skype (composite-index timestamps), Google (protobuf blobs), Camera / Docs / Drive / Gmail / Duo / Maps (no conventional local store).

5. Known limitations

1. **Match inflation.** Ground truth is matched to events within a 5-minute window, and one event can satisfy multiple GT rows. TikTok records 8 matches from 6 events; WhatsApp 8 from 7. Strict one-to-one matching would reduce 68.2%.
 2. **Heuristic epoch detection.** 14,749 of 30,609 events used magnitude-based format detection rather than a declared format. Reliable in the observed range but not guaranteed at the edges.
 3. **Single-device corpus.** No cross-device or cross-version validation.
 4. **INDIRECT_ONLY untested.** 77 Android Auto events (~25 commutes) have no direct artifact and must be inferred from Bluetooth pairing, power state and app-usage correlation. This is the most interesting reasoning task in the dataset and is currently unevaluated.
 5. **Small labelled set.** 28 pairs.
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6. Evaluation plan

Score model output on the held-out set against three measures:

- **Recall** — proportion of ground truth events reconstructed
- **Citation validity** — proportion of emitted EVT-IDs that resolve to a real event, and whose content matches that event
- **Absence handling** — proportion of unrecoverable events correctly reported as unrecoverable rather than silently omitted or fabricated

The second is the anti-hallucination measure and is fully automatable: parse IDs from output, look each up in `events.jsonl`, compare.

7. Pipeline (for reproduction)

#	Stage	Output
1	Android 9 logical image	7.46 GB decrypted /data partition
2	<code>inventory_sqlite.py</code>	576 databases, 1,086 non-empty tables

#	Stage	Output
3	<code>profile_artifacts.py</code>	schemas for 30 target packages
4	<code>extract_events.py</code> + <code>specs_extra.py</code> + <code>specs_extra2.py</code>	27 artifact specs
5	<code>events.jsonl</code>	30,609 provenance-linked events
6	<code>ground_truth.csv</code>	322 events from image documentation
7	<code>build_llm_dataset_v2.py</code>	windowing and target generation
8	<code>train.jsonl</code> / <code>unlabelled.jsonl</code>	28 pairs / 52 windows
9	<code>evaluate.py</code>	baseline metrics
10	Fine-tuning	<i>current stage</i>

All extraction code is standard-library Python. No commercial forensic tools were used at any stage.